Extending Imperfect Retraining Analysis to Three-Component Machine Learning Systems

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I. Overview

Prior work established that retraining a component in a two-component machine learning system (MLS) does not guarantee system-level improvement, a phenomenon termed *entangled enhancement*, and modeled the resulting availability-continuity trade-off using a Semi-Markov Process (SMP). However, real-world MLS commonly consist of three or more sequentially dependent components, a case that remains entirely unmodeled. This proposal defines a research agenda to extend imperfect retraining analysis to three-component MLS, grounded in the autonomous driving perception-prediction-planning pipeline using the nuScenes dataset. We identify three new failure modes unique to three-component chains, propose four research questions, and outline a nine-month methodology combining controlled retraining experiments with an extended SMP model calibrated from real data, directly addressing the empirical validation gap of the prior work.

II. Introduction

Modern intelligent systems increasingly rely on multi-component Machine Learning Systems (MLS), where the output of one model directly feeds into the next. A critical operational challenge arises when such systems require *retraining* to adapt to distributional shifts, a process that is rarely perfect in practice.

Wang and Machida [1] formalized this as *imperfect retraining*: a retraining event that improves a component's individual performance does not guarantee improvement in overall system-level performance. This phenomenon, termed **entangled enhancement**, was modeled for two-component MLS using a Semi-Markov Process (SMP), revealing a fundamental trade-off between *system availability* and *service continuity*.

However, the prior work explicitly acknowledges a critical open problem: real-world MLS commonly consist of **three or more** sequentially dependent components, such as the backbone $\rightarrow$ neck $\rightarrow$ head architecture in modern object detectors, or the perception $\rightarrow$ prediction $\rightarrow$ planning pipeline in autonomous driving. Neither the theoretical modeling nor the empirical behavior of entangled enhancement in such systems has been studied.

ISSRE 2025 reviewers independently raised this gap, alongside concerns about synthetic parameters and lack of empirical validation. This proposal defines a research agenda that addresses all three concerns simultaneously by extending imperfect retraining analysis to three-component MLS with real-world experimental grounding.

III. Motivation and Problem Statement

A. From Two to Three Components

In a two-component system $C_1 \rightarrow C_2$, entangled enhancement is pairwise: retraining C_1 may degrade C_2 if their performance spaces are coupled. The existing SMP model captures this with a four-state system performance matrix [1].

In a three-component system $C_1 \rightarrow C_2 \rightarrow C_3$, three fundamentally new failure modes arise that are invisible in two-component analysis:

1) **Cascade degradation:** Retraining C_1 degrades C_2, which *in turn* degrades C_3, a second-order failure that propagates silently through the chain.
2) **Partial recovery ambiguity:** If C_1 and C_3 improve but C_2 degrades, the system state is undefined under existing binary performance thresholds.
3) **Policy complexity:** The choice of which component to retrain, and in what order, introduces a combinatorial decision space, $2^3 - 1 = 7$ non-trivial subsets, absent in two-component systems.

These are qualitatively new behaviors requiring new modeling and new empirical grounding.

B. Why the Prior SMP is Insufficient

The existing SMP has $2^2 = 4$ states for two components. A naive extension to $2^3 = 8$ states is necessary but not sufficient: it does not capture the directional propagation of representation shift between components, nor the asymmetric coupling that emerges when an intermediate component (C_2) both receives and transmits degraded features. A structural extension to the transition model is required.

IV. Research Questions

RQ1 (Characterization): How does entangled enhancement manifest in three-component sequential MLS, and what new failure modes emerge that are absent in two-component systems?

RQ2 (Modeling): Can the Semi-Markov Process framework be extended to accurately model system-level availability and continuity in three-component MLS, and what structural modifications are required?

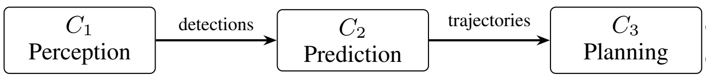
RQ3 (Policy): Given a performance degradation event, what retraining strategy, single-component, ordered multi-component, or full-system retrain, optimally balances availability and continuity in a three-component MLS?

RQ4 (Empirical Grounding): Do theoretically predicted availability-continuity trade-off curves match empirically observed behavior in a real-world three-component MLS under controlled dataset shift?

V. Real-World Use Case

A. Selected System

We ground this research in the three-component autonomous driving pipeline:



Each component corresponds to an independently deployable and retrainable model: C_1 detects objects from raw sensor data (e.g., YOLO11, BEVFormer); C_2 forecasts agent trajectories from detection outputs (e.g., AgentFormer); C_3 produces ego-vehicle paths from predicted trajectories (e.g., PDM-Closed).

B. Justification

Strict sequential dependency. C_2 operates on bounding boxes from C_1; C_3 operates on trajectory forecasts from C_2. Any representation shift propagates directly downstream, exactly the condition for entangled enhancement.

Independent retrainability. Each component has a well-defined, independently measurable performance metric: mAP for perception, minADE/minFDE for prediction, and collision rate or L2 displacement error for planning.

Natural dataset shift. Domain shifts occur organically, day$\rightarrow$night, clear$\rightarrow$rain, city-to-city, enabling realistic, reproducible retraining experiments without synthetic parameters.

Safety-critical stakes. The availability-continuity trade-off has direct safety implications in autonomous driving, amplifying the real-world relevance of findings.

Continuity with prior work. The original paper used the KITTI autonomous driving dataset [1]; this proposal deepens the same domain thread.

C. Dataset

Primary: nuScenes [2], 1,000 driving scenes, multi-modal (camera + LiDAR), annotated for detection, tracking, and prediction across Boston and Singapore, providing natural geographic domain shift.

Secondary: KITTI [3] for backward comparability with prior work; Waymo Open Dataset [4] for scale validation.

VI. Methodology

A. Phase 1, Empirical Setup

Baseline: Train C_1, C_2, C_3 independently on the nuScenes Boston split. Record per-component metrics and the end-to-end system metric (planning L2 displacement error).

Dataset shift injection: Introduce the nuScenes Singapore split as the target domain. This operationalizes a realistic distributional shift with known structure.

Controlled retraining experiments: Retrain each of the 7 non-trivial component subsets $\{C_1\}, \{C_2\}, \{C_3\}, \{C_1, C_2\}, \{C_1, C_3\}, \{C_2, C_3\}, \{C_1, C_2, C_3\}$ and measure system-level impact after each.

Entanglement measurement: For each experiment, record cases where a component's individual metric improves but the system-level metric does not, directly operationalizing entangled enhancement.

B. Phase 2, Theoretical Extension

State space extension: Formalize the 8-state SMP for three components. Each state $\langle s_1, s_2, s_3 \rangle \in \{0, 1\}^3$ encodes whether each component is above its performance threshold.

Cascade transition modeling: Introduce second-order transitions, state changes in C_3 triggered not by direct retraining but by the representation shift propagated from C_1 retraining through C_2. This is the novel structural contribution absent from the prior SMP.

Metric extension: Define a *weighted system performance score* to resolve the partial recovery ambiguity, where not all components degrade or improve simultaneously.

Parameter calibration: Derive all SMP transition rates from Phase 1 experimental data, replacing the synthetic parameters of the prior work.

C. Phase 3, Policy Analysis

Policy enumeration: Using the calibrated SMP, evaluate all retraining decision policies, rules mapping current system state to a retraining action.

Trade-off surface: Generate the availability-continuity trade-off surface as a function of per-component retraining frequency and identify the Pareto-optimal frontier.

Sensitivity analysis: Assess how trade-off curves shift under varying shift severity, retraining cost, and component coupling strength.

D. Edge Cases Addressed

Table I summarizes how the methodology handles known risks.

TABLE I
Edge Cases and Mitigations

Edge Case	Mitigation
Middle component non-evaluability	C_2 (prediction) has direct independent metrics (minADE/FDE); no unevaluable stage
Partial system improvement	Weighted system performance score resolves binary ambiguity
Retraining order dependency	Phase 3 explicitly compares sequential vs. simultaneous policies
Synthetic parameters	All SMP rates derived from nuScenes experiments (Phase 2)
2-component backward compatibility	3-component SMP degenerates to prior model when $C_2 \equiv C_3$
Simultaneous multi-component drift	Phase 1 includes multi-component shift scenarios

VII. Expected Contributions

C1: Failure taxonomy: First formal characterization of cascade entangled enhancement in three-component MLS, identifying failure modes unique to chains of length ≥ 3.

C2: Extended SMP model: A three-component Semi-Markov Process with empirically calibrated parameters, the first of its kind and a direct response to the open problem stated in [1].

C3: Retraining policy guide: Concrete, actionable retraining strategies with quantified availability-continuity trade-off surfaces for three-component MLS operators.

C4: Reproducible benchmark: A public experimental benchmark on nuScenes for imperfect retraining in sequential MLS, enabling future replication and extension by the community.

C5: Empirical validation: Real parameters from real data replace the synthetic assumptions that limited the prior work's practical impact and reviewer confidence.

VIII. Conclusion

The ISSRE 2025 paper established a sound theoretical framework for imperfect retraining in two-component MLS, but its own reviewers identified three limitations: only two components are covered, parameters are synthetic, and there is no empirical validation. This proposal directly answers all three.

By targeting the autonomous driving perception-prediction-planning pipeline on nuScenes, we create the conditions for a stronger theoretical model and genuine empirical grounding simultaneously. The research questions are tightly scoped, the methodology is executable within a single research cycle, and the five contributions are concrete and novel.

This work is not merely an incremental extension. Cascade entangled enhancement, where a single retraining event silently propagates degradation through multiple downstream components, is a qualitatively new phenomenon with direct implications for the reliability of safety-critical intelligent systems. Modeling and mitigating it is a necessary step from a promising two-component framework to a field-defining body of work on multi-component MLS reliability.

References

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